

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.366282793	delta=+0.0146852
ratio=1.0062448		
CoG x [m]	-0.002024708562 -> -0.002658739924	delta=-0.000634031
ratio=-		
CoG y [m]	-3.052657894e-05 -> 0.0008627471103	delta=+0.000893274
ratio=-		
CoG z [m]	0.009509749599 -> 0.01046697945	delta=+0.00095723
ratio=-		
rotor effectiveness 1	1 -> 0.9624033902	delta=-0.0375966 ratio=0.96240339
rotor effectiveness 2	1 -> 0.973403579	delta=-0.0265964 ratio=0.97340358
rotor effectiveness 3	1 -> 0.9592877675	delta=-0.0407122 ratio=0.95928777
rotor effectiveness 4	1 -> 0.9518164799	delta=-0.0481835 ratio=0.95181648
rotor command lag [s]	0.004997730255 -> 0.1953933639	delta=+0.190396
ratio=-		

Inertia matrix [kg m^2]

reference:

```
[ 0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[ 1.901508003e-05, 5.9167305e-08, 0.1289921107]
```

estimated:

```
[ 0.06501171244, -1.001772766e-05, 2.318071564e-05]
[-1.001772766e-05, 0.06524204592, 6.925927988e-06]
[ 2.318071564e-05, 6.925927988e-06, 0.12899475]
```

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516 -> 0.06501126897	delta=+5.86238e-05
ratio=1.0009026		
principal moment 2 [kg m^2]	0.06500006701 -> 0.06524248025	delta=+0.000242413
ratio=1.0037294		
principal moment 3 [kg m^2]	0.1289921163 -> 0.1289947591	delta=+2.64282e-06
ratio=1.0000205		

Optimization

joint dynamics loss: 1.1434965334
Gaussian-prior cost: 0.36039351331
selected strict-ZOH rotor lag [s]: 0.195393363886

Bag failure_1_single

reconstruction position RMSE [m]: 0.001565600907
reconstruction position component RMSE [m]: [0.000908 0.000474 0.001184]
reconstruction orientation RMSE [deg]: 0.845162170593
gyro consistency RMSE xyz [rad/s]: [0.051248 0.102544 0.031019]
specific-force consistency RMSE xyz [m/s^2]: [0.946137 0.147336 0.278247]
inferred external force RMS xyz [N]: [0.828218 0.390485 3.05836]
inferred external torque RMS xyz [N m]: [0.076453 0.244106 0.098273]
bag dynamics loss: 1.1434965334
gimbal command lag [s] 0.004998683929 -> 0.03932730441

Recorded command timestamp intervals (data-derived)

Bag failure_1_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s

gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s

Selected optimizer diagnostics

status=1 success=True nfev=4 njev=4
message='gtol' termination condition is satisfied.
reported optimality=5.775235756289021e-08
initial cost=1.5038903291662329
final cost=1.5038900847290744
cost reduction=2.44437158514188e-07
coordinate step L2=0.0002752809049669324
final gradient inf-norm=5.775235756289021e-08
gradient tolerance satisfied=True
final Jacobian rank=14
final Jacobian cond(nonzero)=410.4150611665243
max directional finite-difference Jacobian relative error=8.195483990518276e-12